Artificial Intelligence and Machine Learning

Neural Networks

Lecture Outline

- Classification Metrics
- Neural Networks
  - Forward pass
  - Backward pass

Classification Metrics

- **Accuracy:**

  Accuracy measures the proportion of correctly classified samples out of the total number of samples.

$$Accuracy = \frac{Correct\ Samples}{All\ Sample}$$

99%

198/200 = 0.99

Question: Why not optimizing for the accuracy directly instead of using LogLoss?
Non-differentiable

Could also be written as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

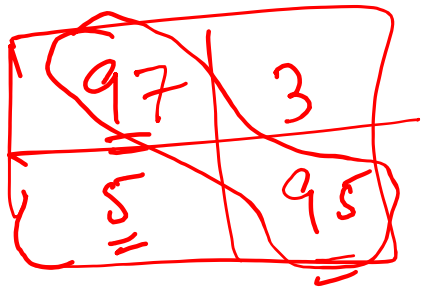
Where:

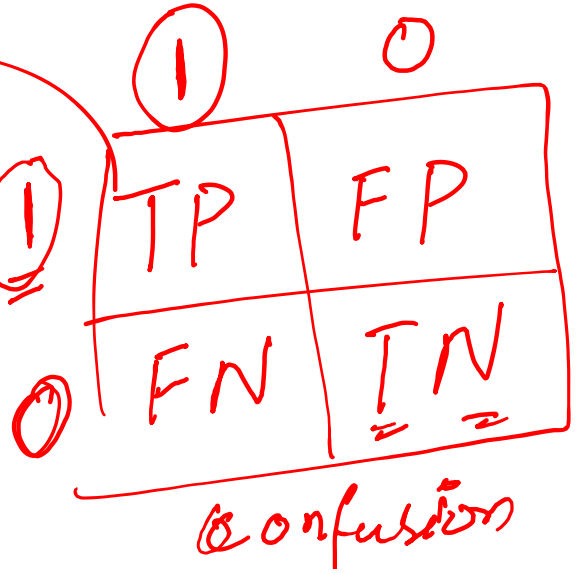
TP: True Positives

TN: True Negatives

FP: False Positives

FN: False Negatives



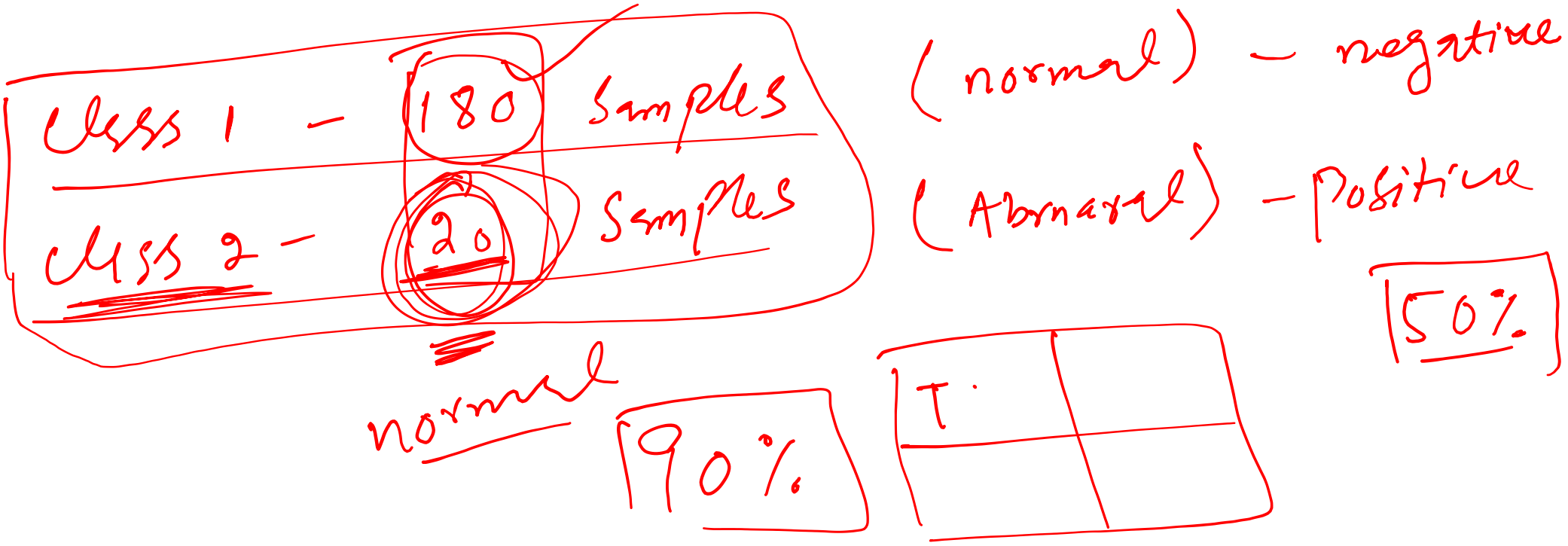


Classification Metrics

But accuracy isn't always enough!

Imagine an imbalanced dataset with 95% negative labels (e.g., "not spam").

A model predicting "negative" all the time will be 95% accurate but useless.



Classification Metrics

But accuracy isn't always enough!

Imagine an imbalanced dataset with 95% negative labels (e.g., "not spam").
A model predicting "negative" all the time will be 95% accurate but useless.

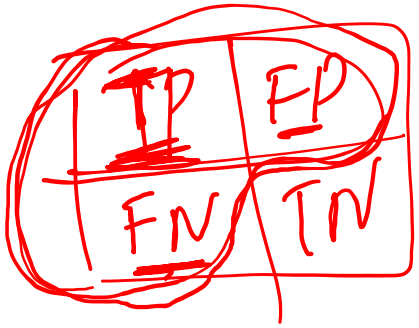
Different goals require different metrics

- Do you care more about **catching all positives** (e.g., cancer detection)?
- Or avoiding **false alarms** (e.g., fraud detection)?

Classification Metrics

- **Precision:**

  Among all the positive predictions the model made, how many are actually correct?
  important when **minimizing false alarms** is critical, like in fraud detection or spam filtering.



$$Precision = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Positives\ (FP)}$$

Classification Metrics

- **Recall:**

  Among all the actual positives, how many did the model correctly identify?

  important when **catching all positives** is vital, such as in cancer detection.

$$Recall = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Negatives\ (FN)}$$

Classification Metrics

- **F1 Score:**

  What if both precision and recall are important?

  Take their (harmonic) mean.

$$F1\ score = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$

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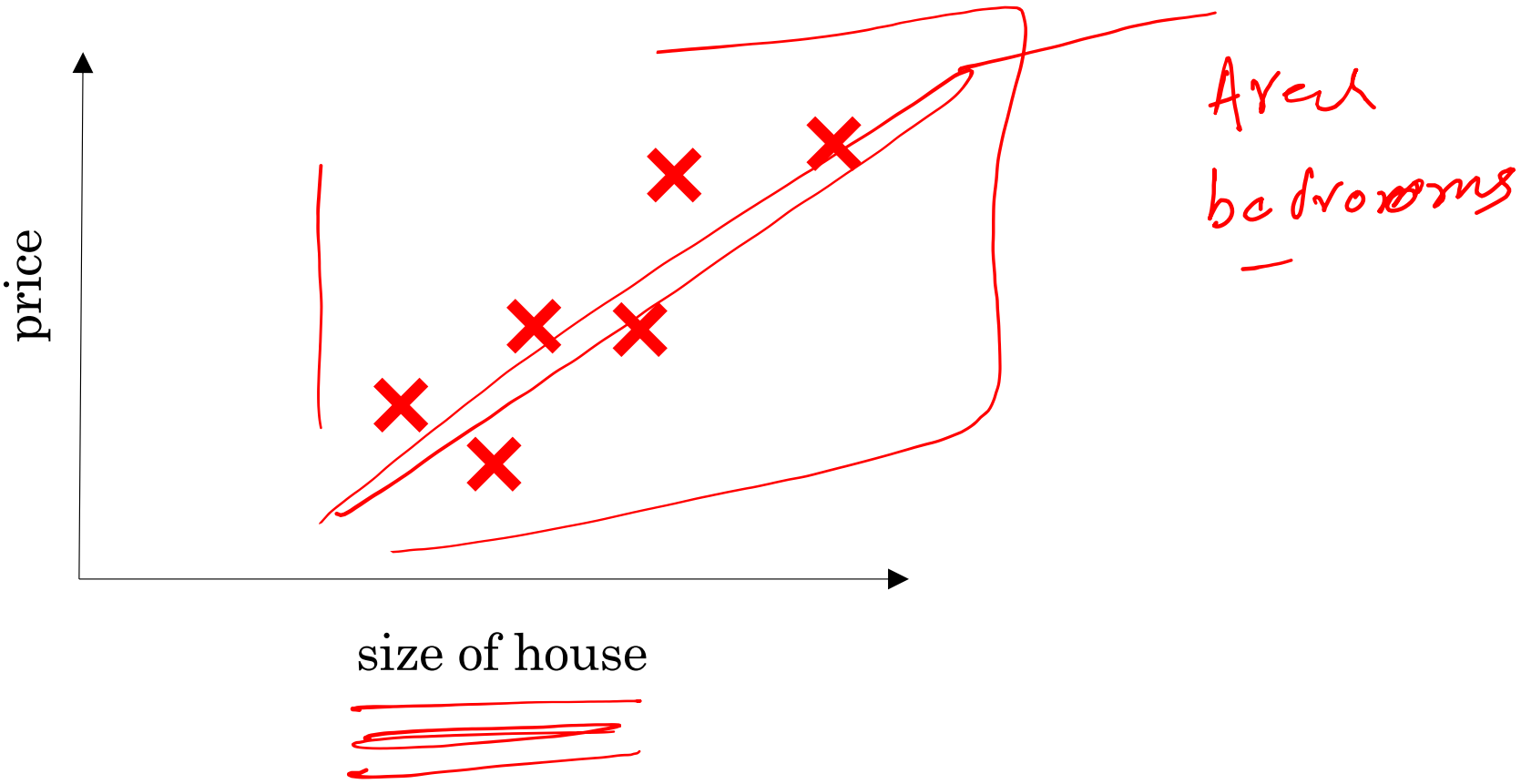
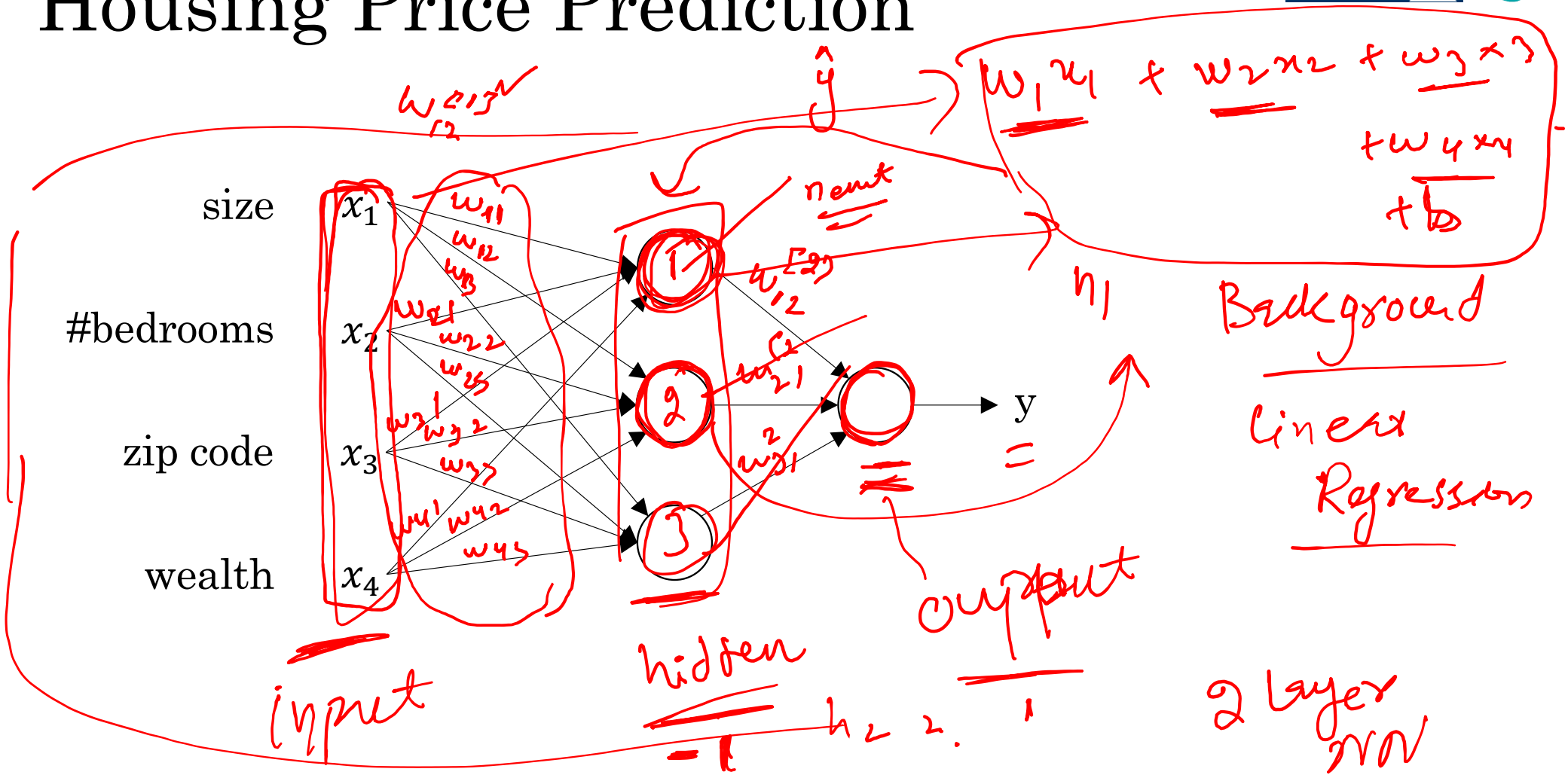
Introduction to Deep Learning

What is a Neural Network?

Housing Price Prediction



Housing Price Prediction



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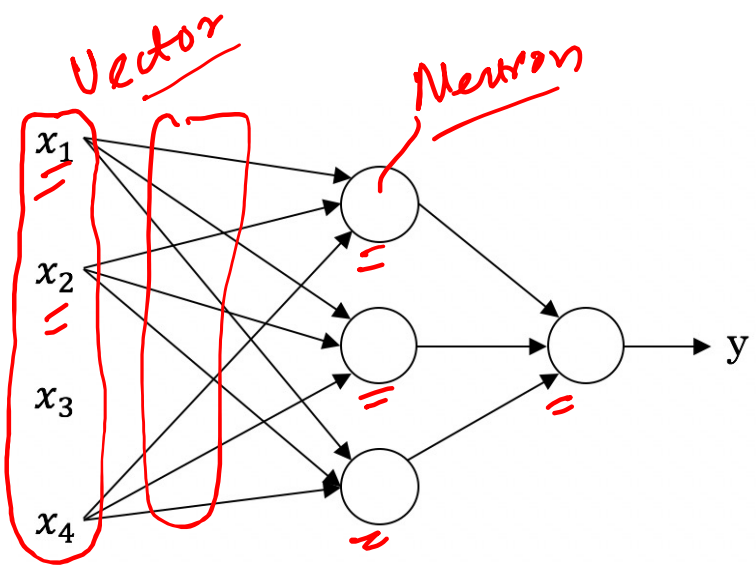
Introduction to Deep Learning

Supervised Learning with Neural Networks

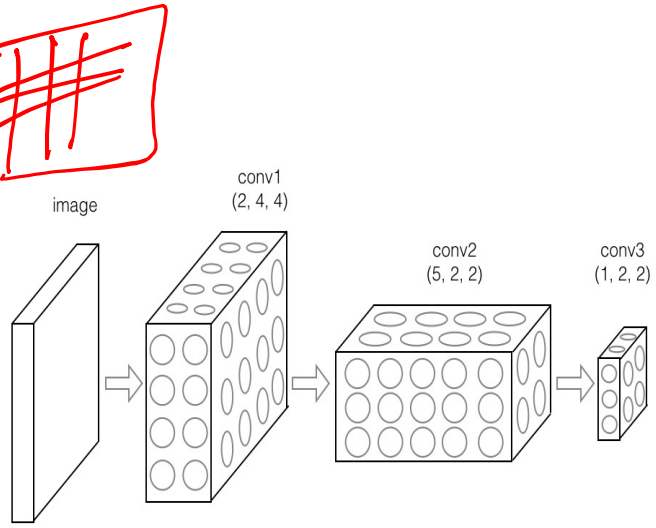
Supervised Learning

Input(x)	Output (y)	Application
Home features	Price	Real Estate
Ad, user info	Click on ad? (0/1)	Online Advertising
Image	Object (1,…,1000)	Photo tagging
Audio	Text transcript	Speech recognition
English	Chinese	Machine translation
Image, Radar info	Position of other cars	Autonomous driving

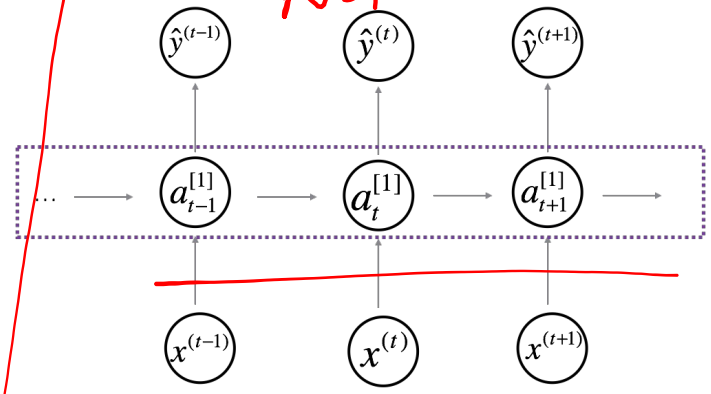
Neural Network examples



Standard NN



Convolutional NN



Recurrent NN

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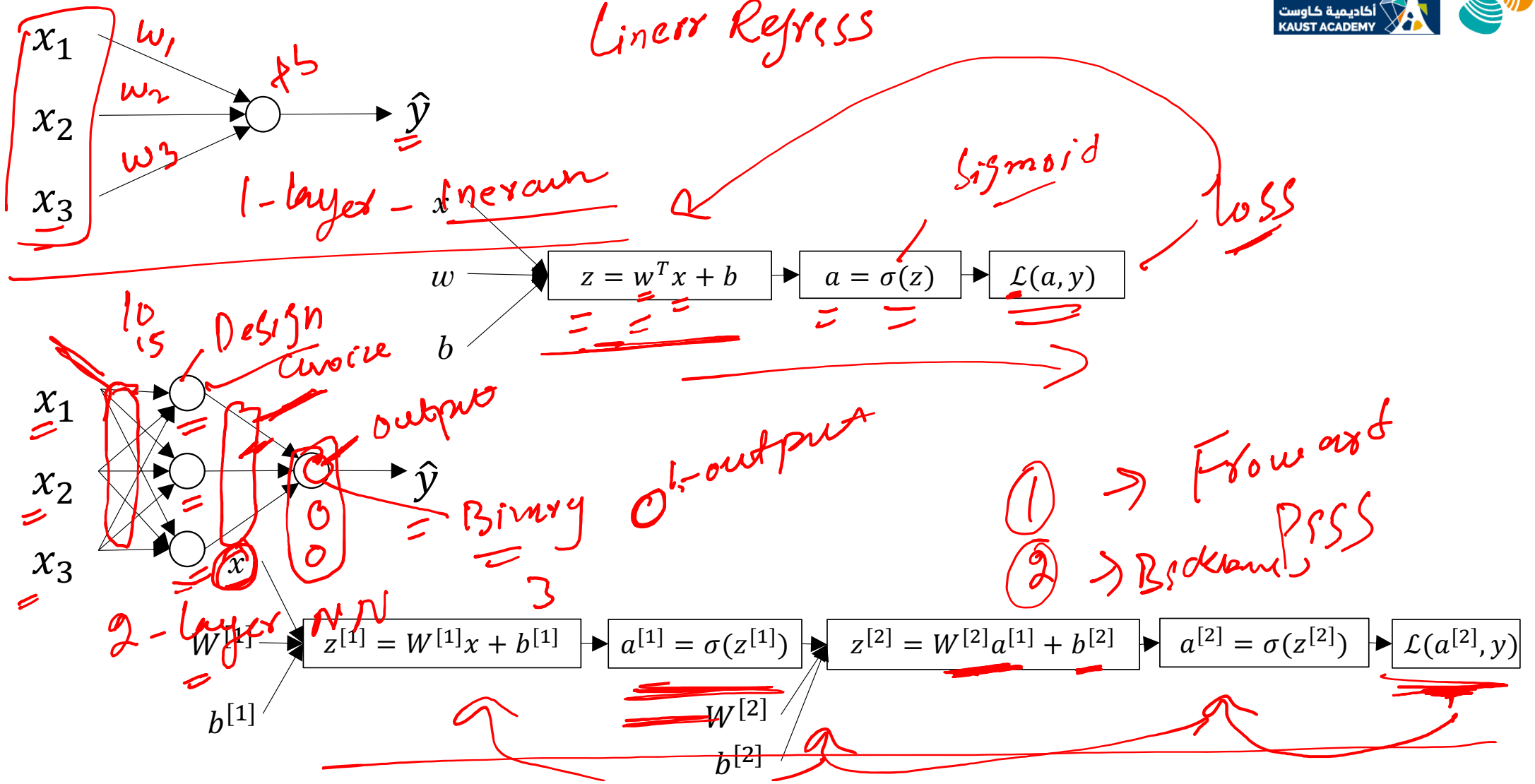
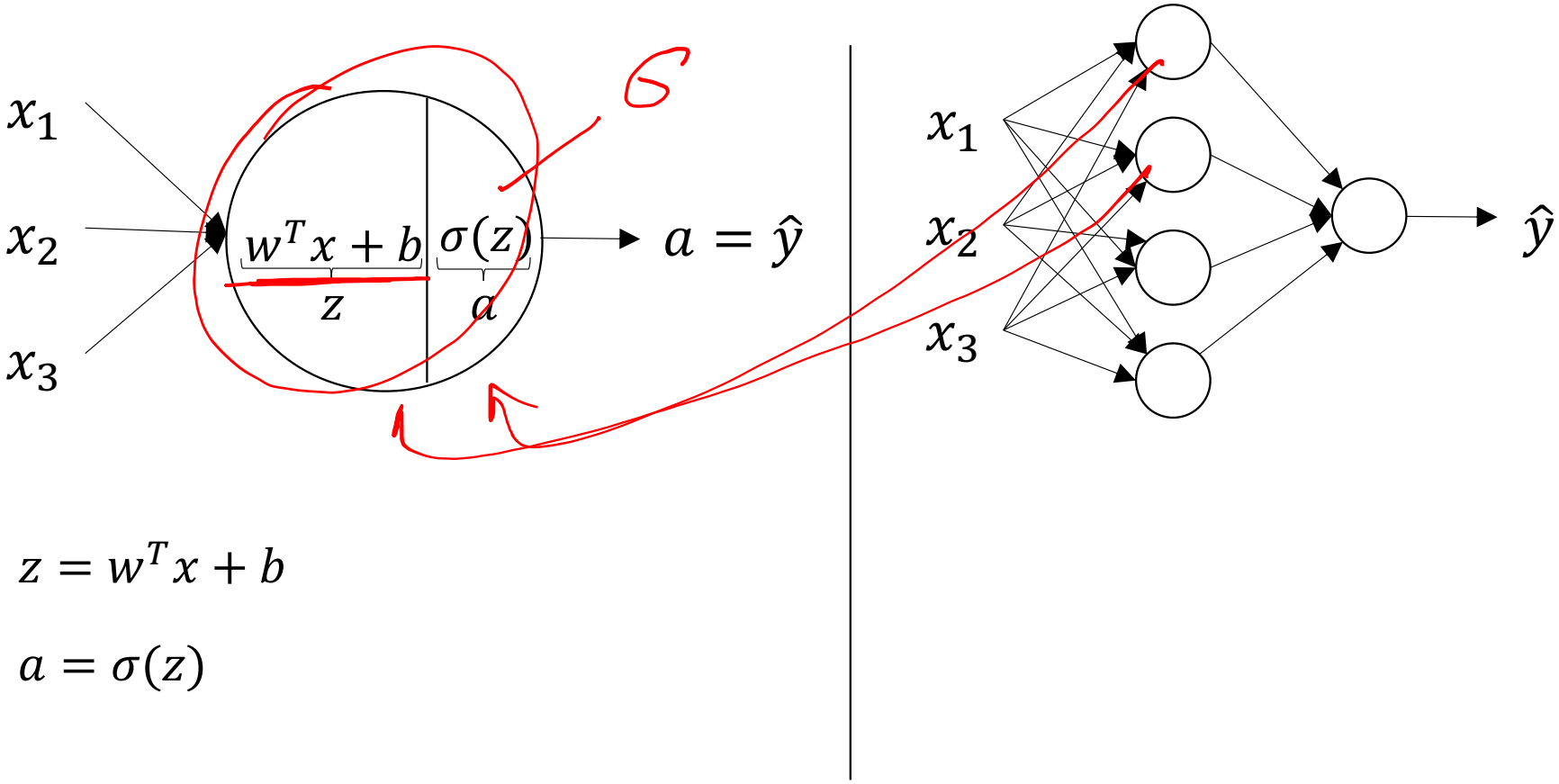
One hidden layer Neural Network

Neural Networks Overview

What is a Neural Network?



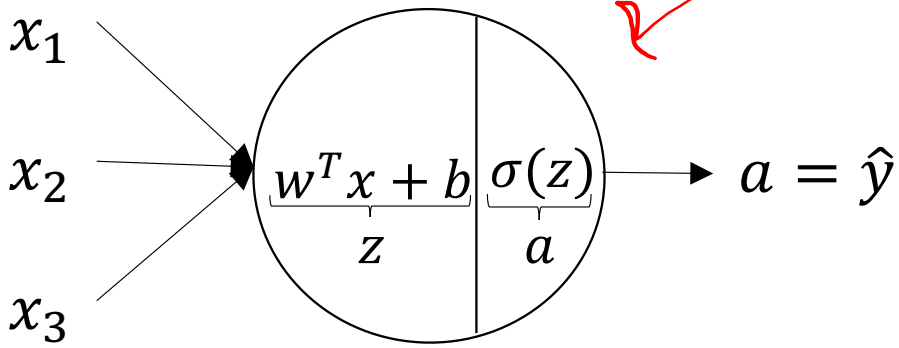
Neural Network Representation



$$z = w^T x + b$$

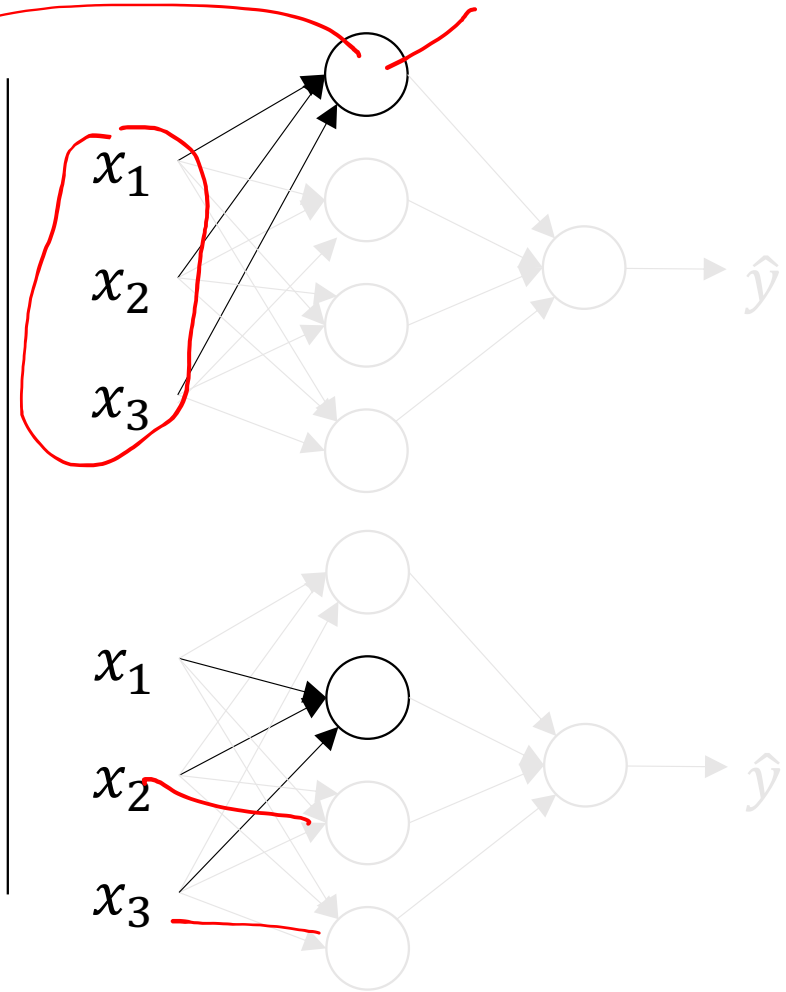
$$a = \sigma(z)$$

Neural Network Representation

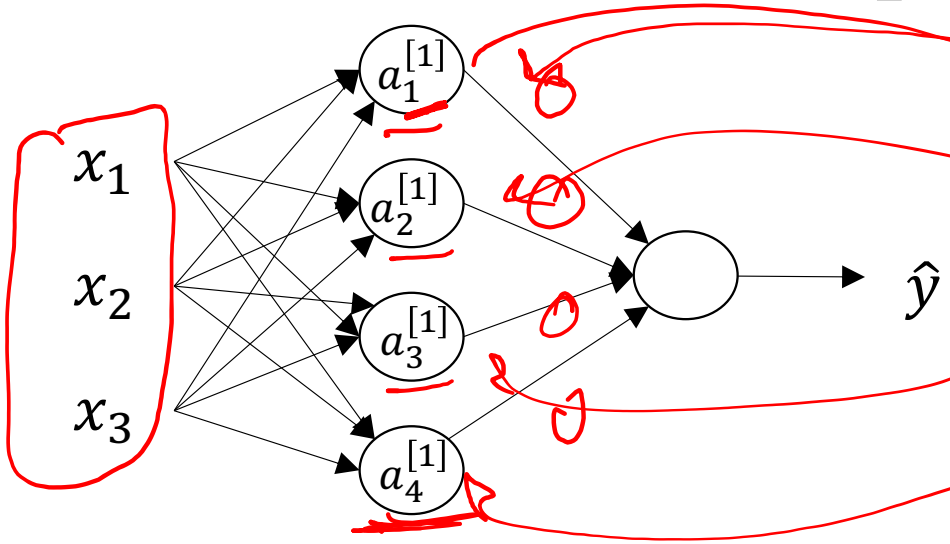


$$z = w^T x + b$$

$$a = \sigma(z)$$



Neural Network Representation



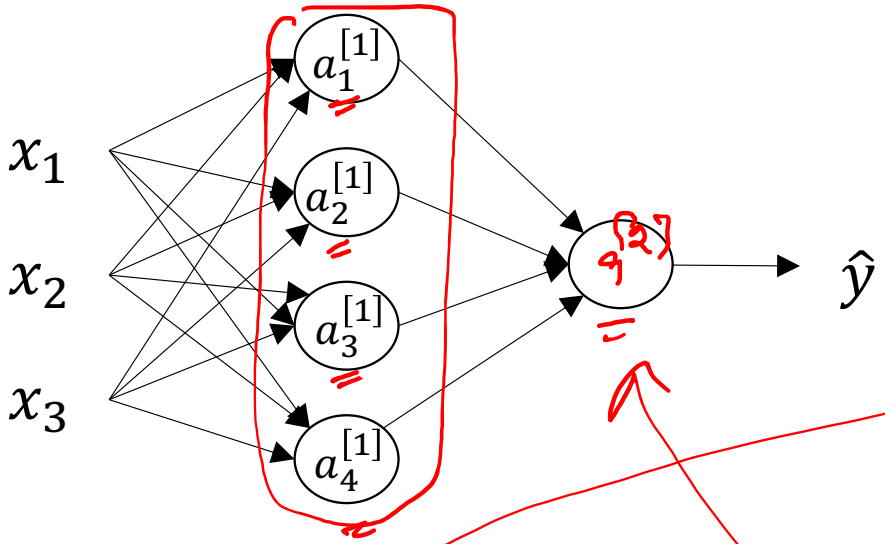
$$z_1^{[1]} = w_1^{[1]T} x + b_1^{[1]}, \quad a_1^{[1]} = \sigma(z_1^{[1]})$$

$$z_2^{[1]} = w_2^{[1]T} x + b_2^{[1]}, \quad a_2^{[1]} = \sigma(z_2^{[1]})$$

$$z_3^{[1]} = w_3^{[1]T} x + b_3^{[1]}, \quad a_3^{[1]} = \sigma(z_3^{[1]})$$

$$z_4^{[1]} = w_4^{[1]T} x + b_4^{[1]}, \quad a_4^{[1]} = \sigma(z_4^{[1]})$$

Neural Network Representation learning



Given input x:

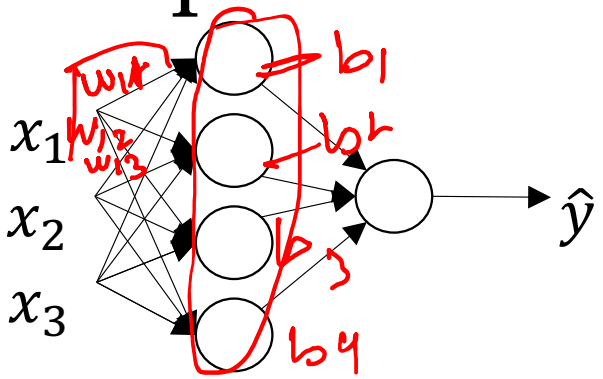
$$z^{[1]} = W^{[1]}x + b^{[1]}$$

$$a^{[1]} = \sigma(z^{[1]})$$

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}$$

$$a^{[2]} = \sigma(z^{[2]})$$

Recap of vectorizing across multiple examples



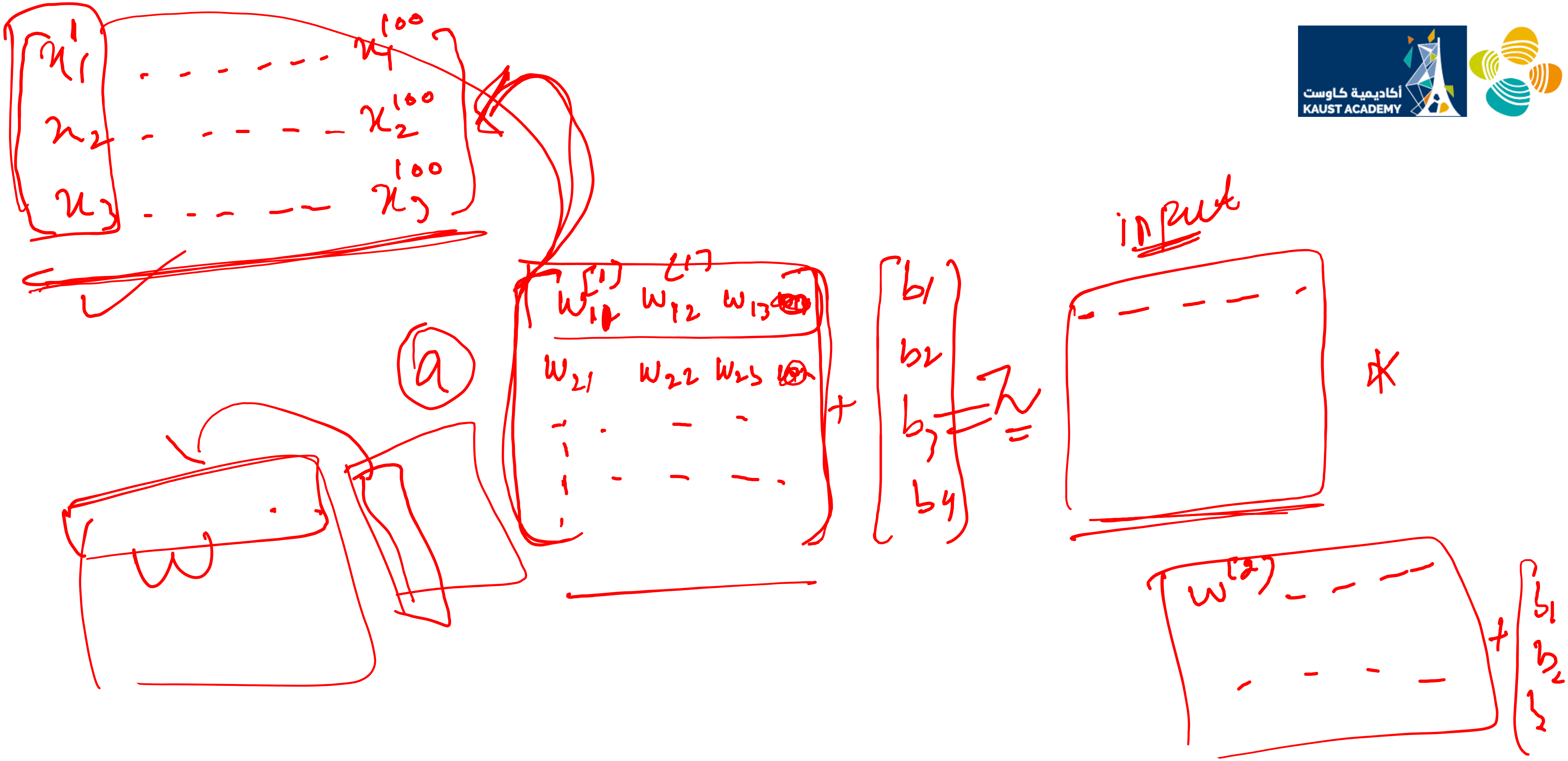
$$X = \begin{bmatrix} | & | & & | \\ x^{(1)} & x^{(2)} & \cdots & x^{(m)} \\ | & | & & | \end{bmatrix}$$

$$A^{[1]} = \begin{bmatrix} | & | & & | \\ a^{[1](1)} & a^{[1](2)} & \cdots & a^{[1](m)} \\ | & | & & | \end{bmatrix}$$

```
for i = 1 to m
```

$$z^{[1](i)} = W^{[1]}x^{(i)} + b^{[1]}$$
$$a^{[1](i)} = \sigma(z^{[1](i)})$$
$$z^{[2](i)} = W^{[2]}a^{[1](i)} + b^{[2]}$$
$$a^{[2](i)} = \sigma(z^{[2](i)})$$

$$Z^{[1]} = W^{[1]}X + b^{[1]}$$
$$A^{[1]} = \sigma(Z^{[1]})$$
$$Z^{[2]} = W^{[2]}A^{[1]} + b^{[2]}$$
$$A^{[2]} = \sigma(Z^{[2]})$$


input

Gradient descent for neural networks

Parameters $w^{[1]}, b^{[1]}, w^{[2]}, b^{[2]}$

$J(w^{[1]}, b^{[1]}, w^{[2]}, b^{[2]})$

$dw^{[1]} = \frac{\partial}{\partial w^{[1]}} J(w^{[1]}, b^{[1]}, w^{[2]}, b^{[2]})$, $db^{[1]} = \frac{\partial}{\partial b^{[1]}} J(w^{[1]}, b^{[1]}, w^{[2]}, b^{[2]})$, ...

$w^{[1]} := w^{[1]} - \alpha\, dw^{[1]}$

$b^{[1]} := b^{[1]} - \alpha\, db^{[1]}$

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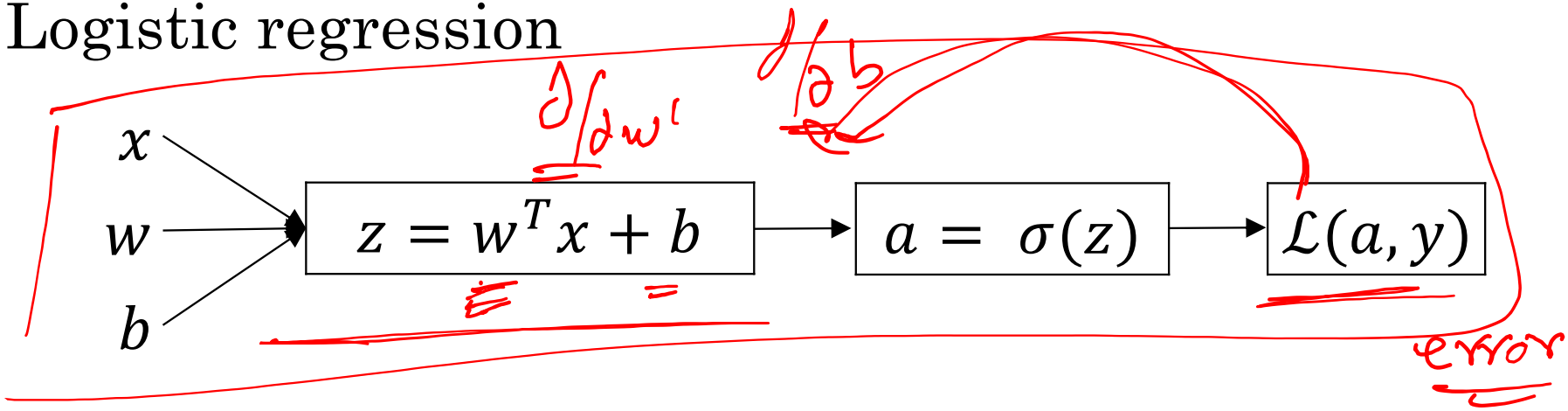
One hidden layer Neural Network

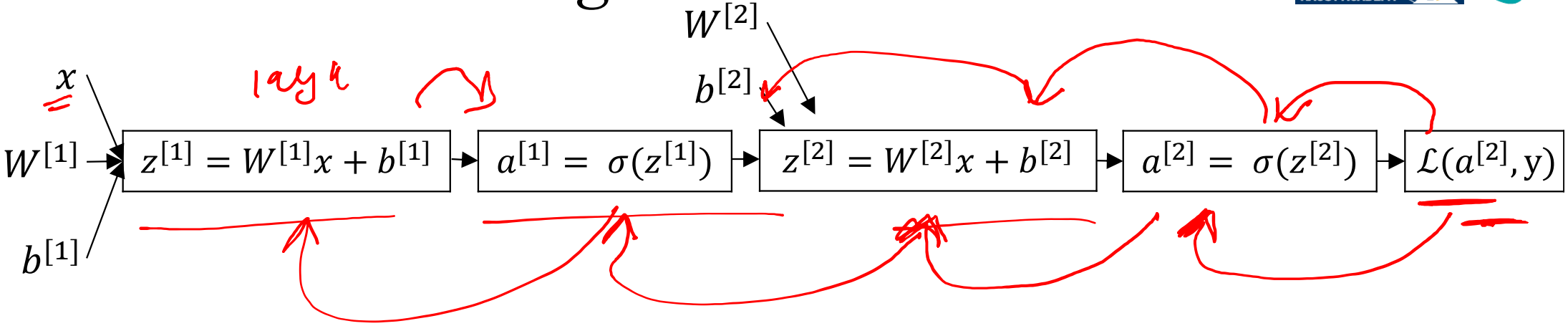
Backpropagation intuition

Computing gradients

Logistic regression



Neural network gradients



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Summary of gradient descent

$$dz^{[2]} = a^{[2]} - y$$

$$dW^{[2]} = dz^{[2]} a^{[1]T}$$

$$db^{[2]} = dz^{[2]}$$

$$dz^{[1]} = W^{[2]T} dz^{[2]} * g^{[1]\prime}(z^{[1]})$$

$$dW^{[1]} = dz^{[1]} x^{T}$$

$$db^{[1]} = dz^{[1]}$$

Andrew Ng

Summary of gradient descent

$$dz^{[2]} = a^{[2]} - y \qquad dZ^{[2]} = A^{[2]} - Y$$

$$dW^{[2]} = dz^{[2]} a^{[1]^T} \qquad dW^{[2]} = \frac{1}{m} dZ^{[2]} A^{[1]^T}$$

$$db^{[2]} = dz^{[2]} \qquad db^{[2]} = \frac{1}{m} np.sum(dZ^{[2]}, axis = 1, keepdims = True)$$

$$dz^{[1]} = W^{[2]T} dz^{[2]} * g^{[1]\prime}(z^{[1]}) \qquad dZ^{[1]} = W^{[2]T} dZ^{[2]} * g^{[1]\prime}(Z^{[1]})$$

$$dW^{[1]} = dz^{[1]} x^T \qquad dW^{[1]} = \frac{1}{m} dZ^{[1]} X^T$$

$$db^{[1]} = dz^{[1]} \qquad db^{[1]} = \frac{1}{m} np.sum(dZ^{[1]}, axis = 1, keepdims = True)$$

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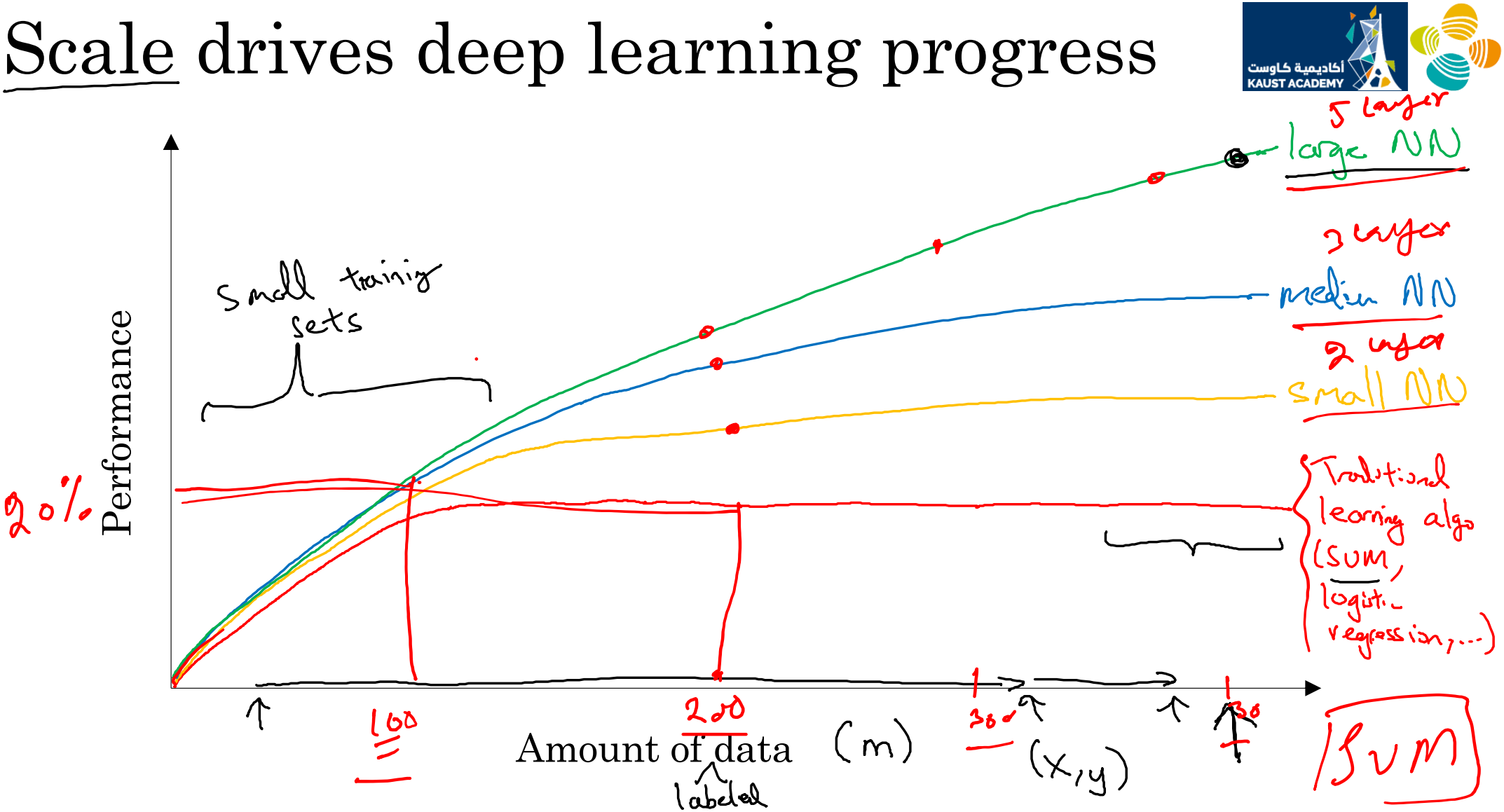
Introduction to Neural Networks

Why is Deep Learning taking off?

Scale drives deep learning progress

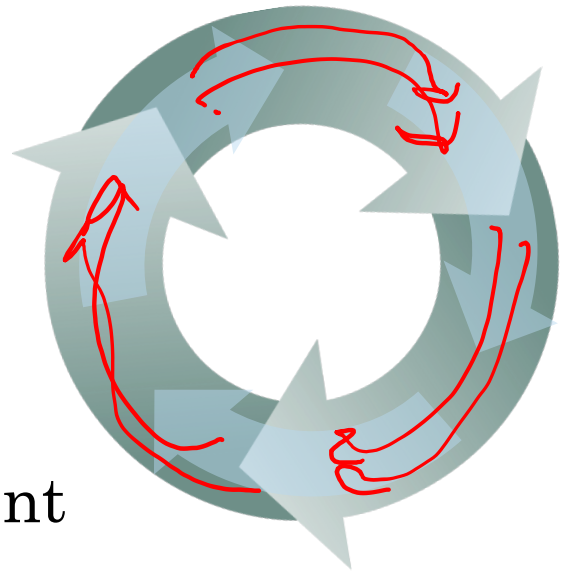


Scale drives deep learning progress

- Data

- Computation

- Algorithms



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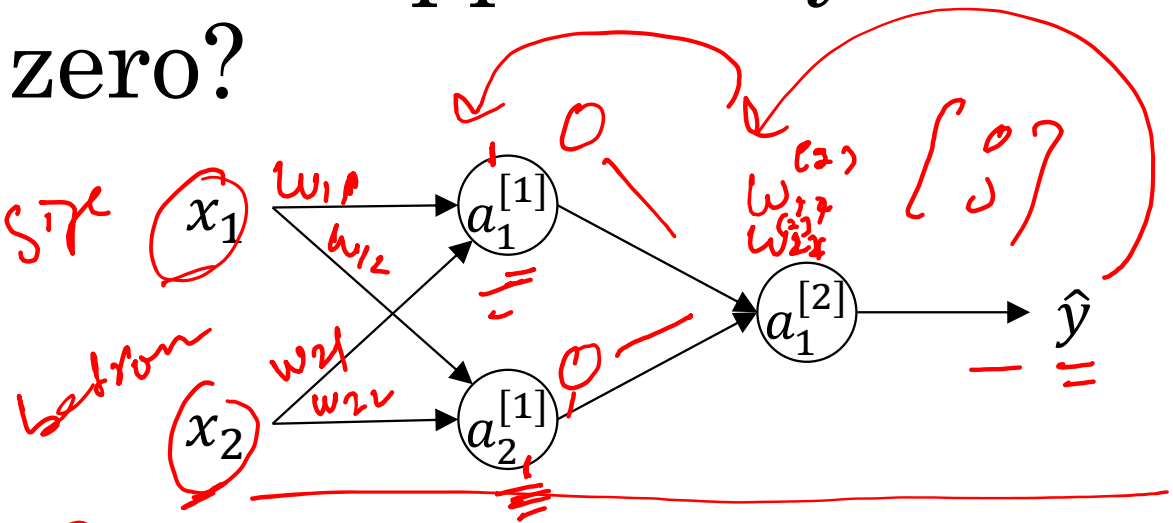
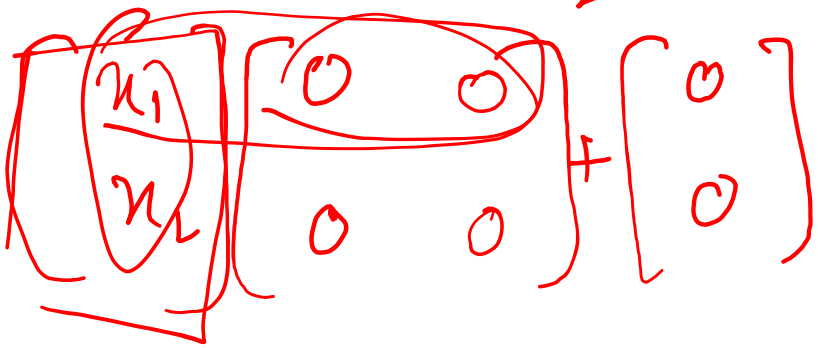
One hidden layer Neural Network

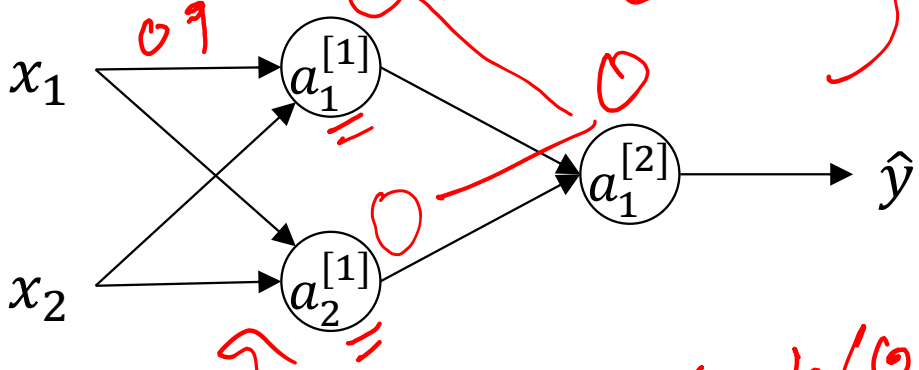
Random Initialization

What happens if you initialize weights to zero?





Random initialization



np.random.randn((2,2)) * 0.01

Better

(0 — 0.01)

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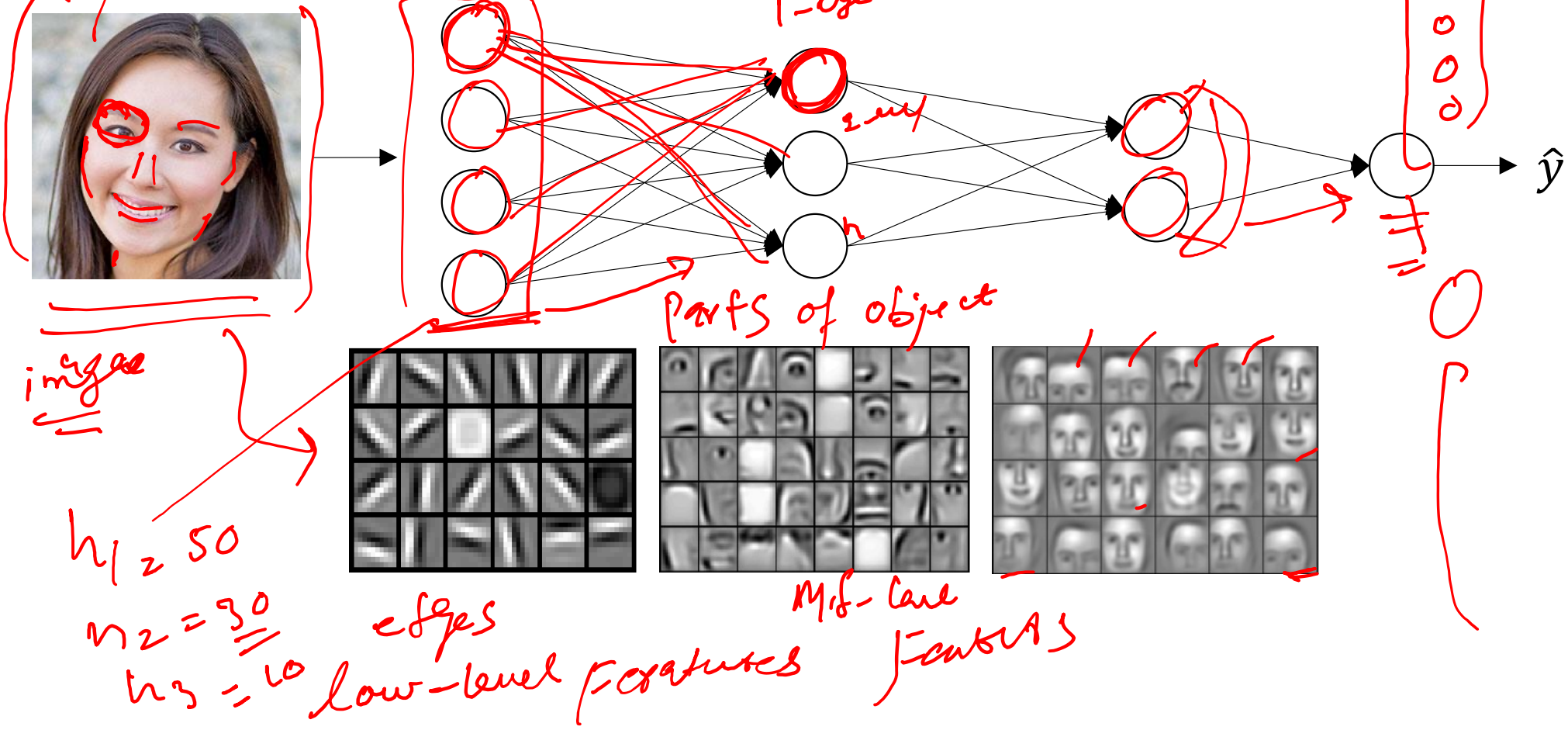
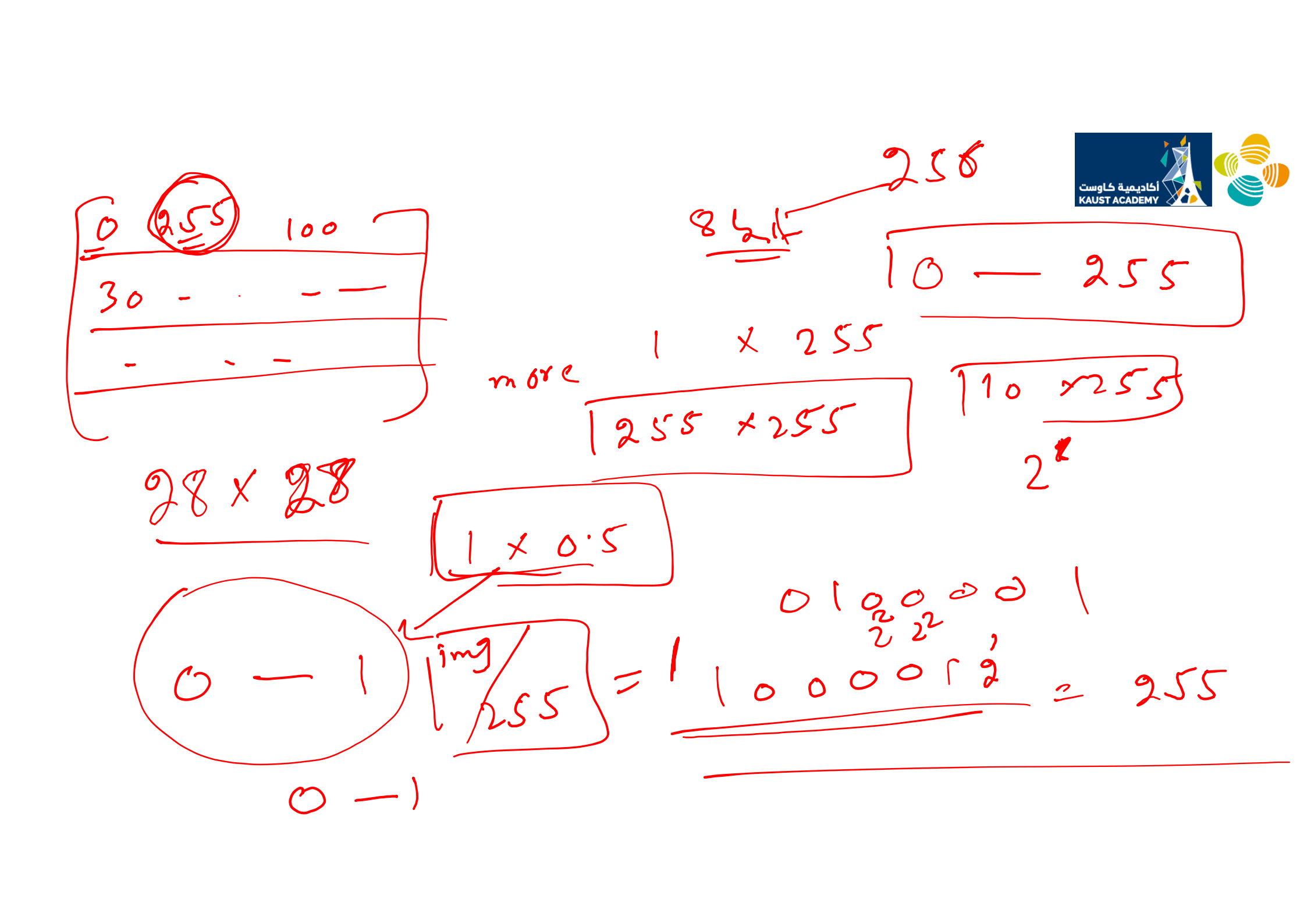
Deep Neural Networks

Why deep representations?

Intuition about deep representation



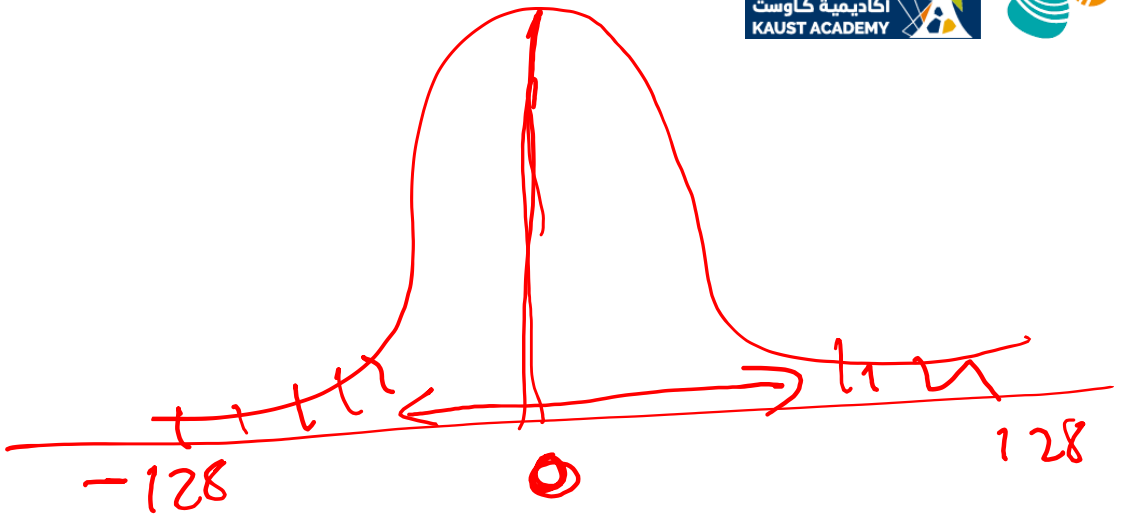

0 255 100
30 - - -
- - -
256
8 bit
0 — 255
more
1 x 255
255 x 255
10 x 255
2^8
28 x 28
1 x 0.5
0 — 1
img
255
= 1
01000001
2 2^2
10000012 = 255
0 — 1

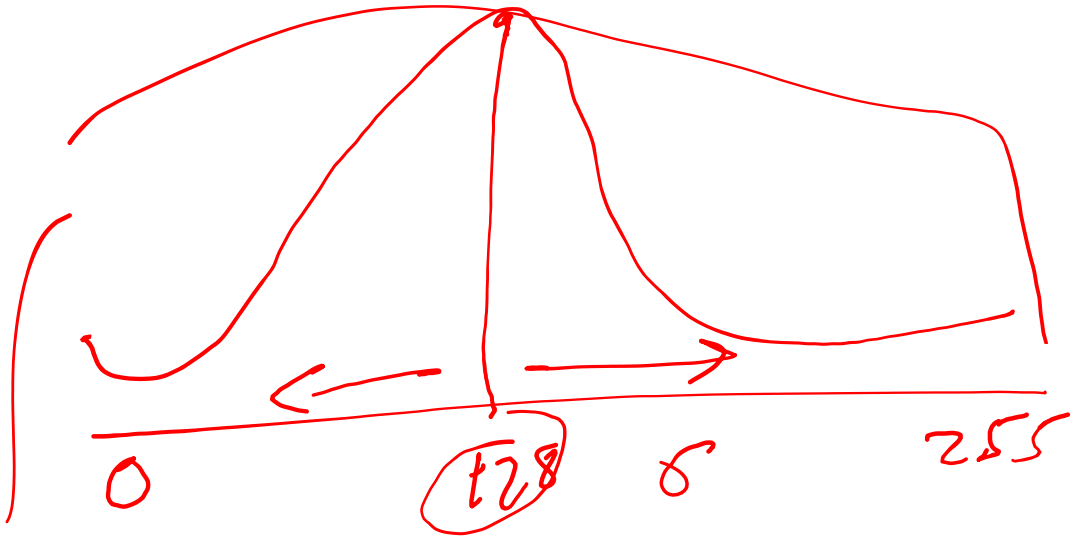
Standardization

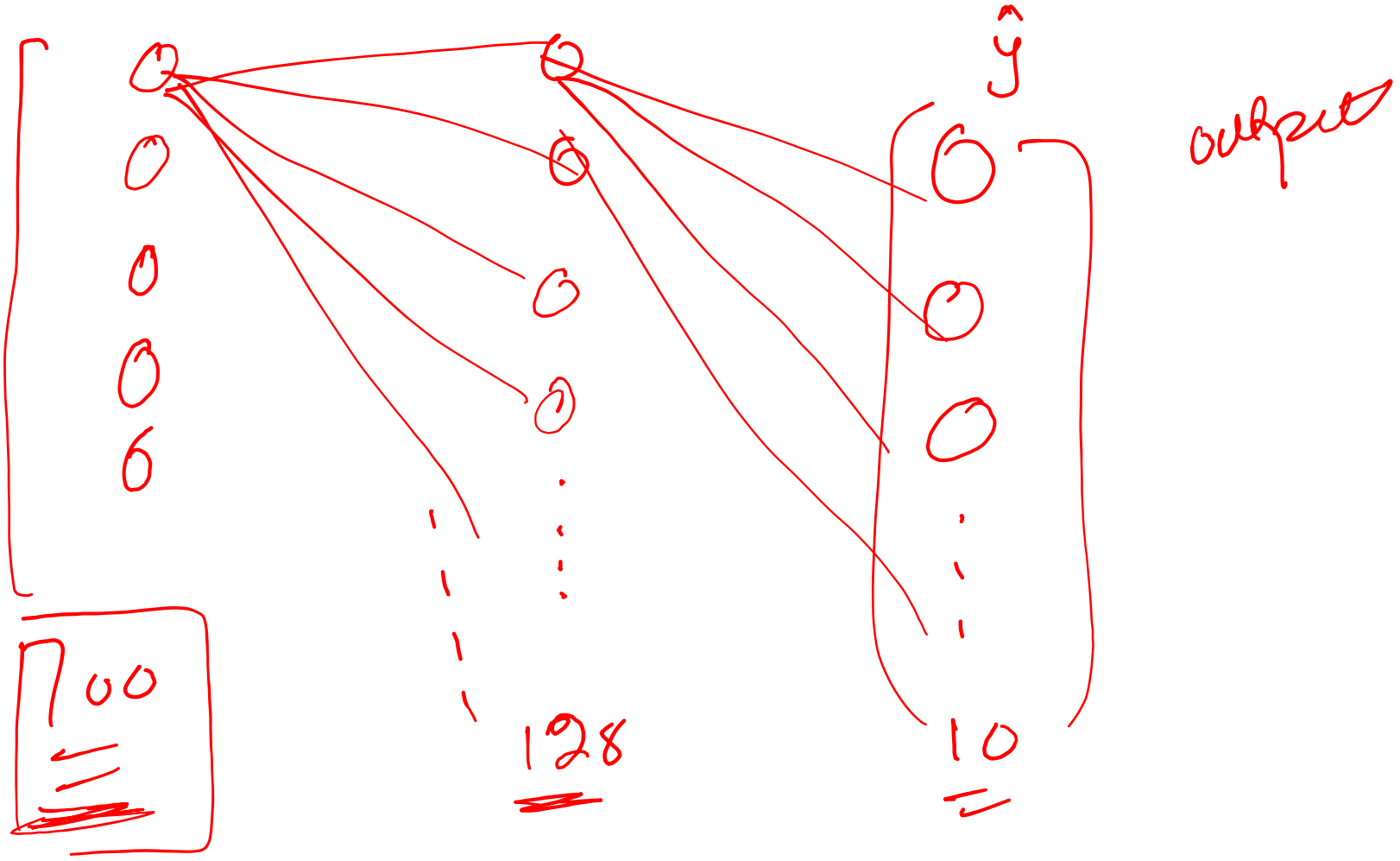
mean = 125

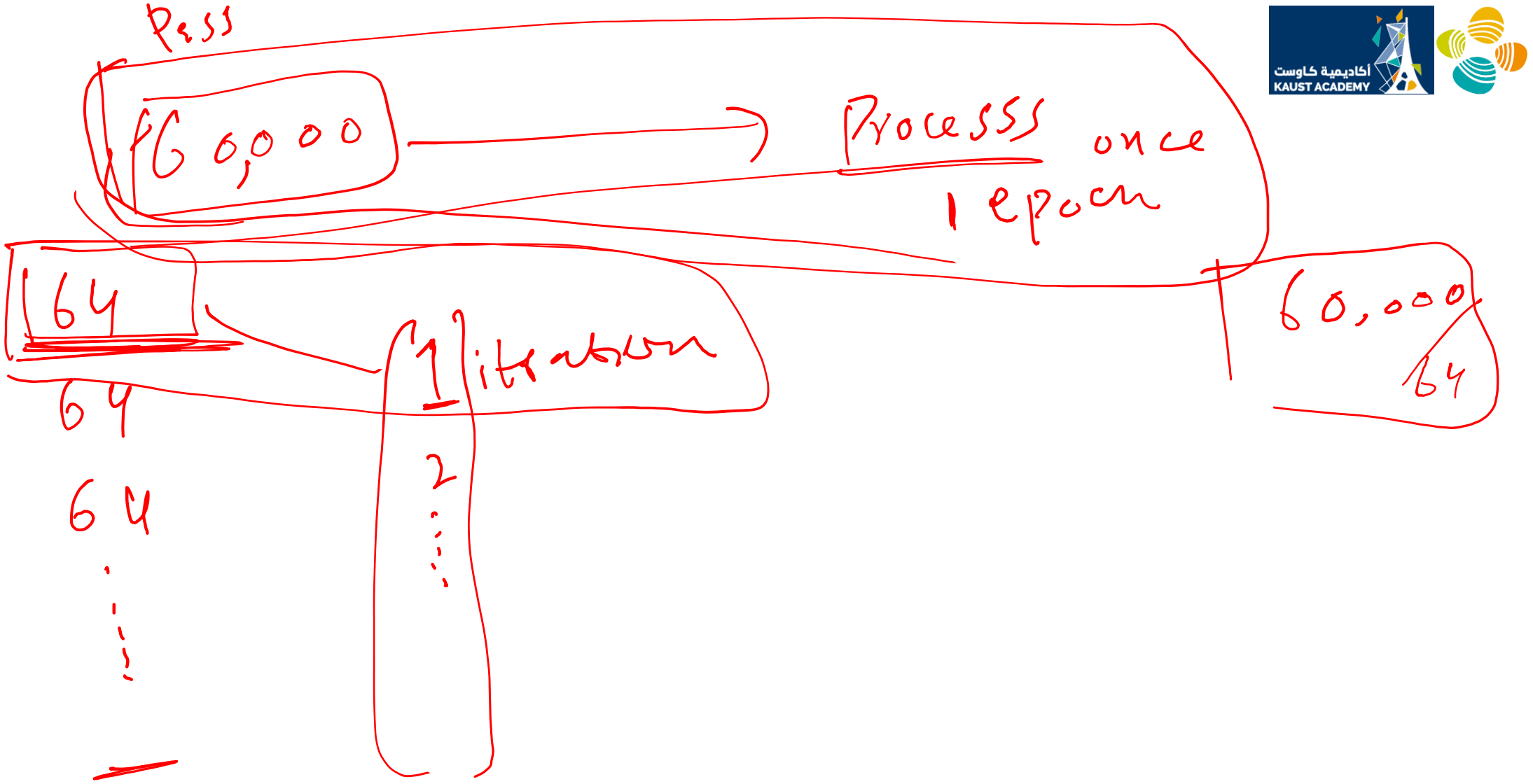
Std = $\sqrt{\quad}$

$$\frac{img(x) - mean}{Std}$$






ŷ
output
784
128
10


Pass
60,000
Process once
1 epoch
64
1 iteration
60,000
64
64
64
2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